

Capitolo 9

Lesson 3-12-2025

9.1 Channel estimation

3 Dec 2025

General idea (pilots) and formal writing of $\bar{Y}$, $\text{vec}(\bar{Y})$

Ref: *A signal processing perspective* (pp. 57-130), p. 54

We've seen OFDM and single carrier equalization. Today professor wants to finish channel estimation. If we want to transmit with OFDM, we need to estimate the channel.

$$y[k] = \sqrt{G}h[k]p[k] + v[k]$$

↑ *pilot component*

$\xrightarrow{\text{IDFT}}$ of length K

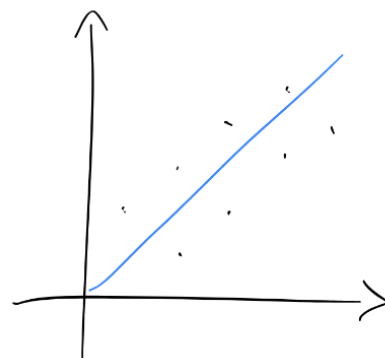
How can we do channel estimation? There are two approaches:

- Maximum likelihood
- Least squares $\rightarrow$ like MD (*Minimum Distance*)

In the book the difference between least square and MMSE is not made clear.

If we have something like this and we're asked to draw a line which minimizes the error we're doing, that is Least Squares (LS).

How do we do channel estimation? We use what we'll call a **pilot**: this is a symbol at a certain frequency and time that we know.



LS minimization visualization

Since we know the symbol we've sent and we know its power,

we can learn the response of the channel accordingly.

Formal writing:

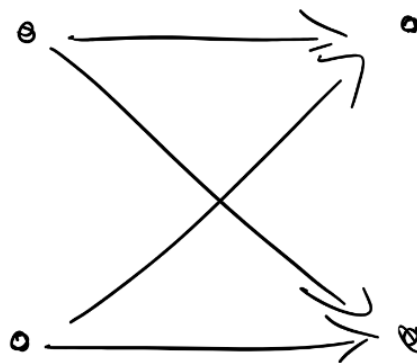
We can model the received signal as

$$y[n] = \sqrt{G} \sum_{\ell=0}^L H[\ell] x[n - \ell] + v[n].$$

$$y[k] = \sqrt{G} \underset{\substack{\uparrow \\ \text{useful}}}{h[k]} p[k] + \underset{\substack{\uparrow \\ \text{noise}}}{v[k]}$$

How the H matrices represent the delay

Let's suppose we have 2 transmit and 2 receive antennas and all the signals arrive without any delay at the receiver:



2x2 MIMO with no delay

We'll use a matrix such as

$$H = \begin{bmatrix} h_{00} & h_{01} \\ h_{10} & h_{11} \end{bmatrix}$$

to characterize the channel.

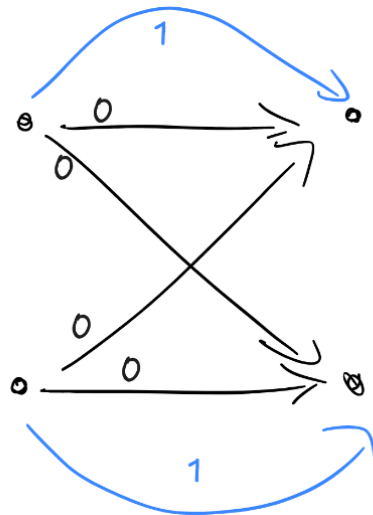
Even in this case, there could be other paths, with different delays!

Here the channel won't be flat anymore (the arrows in blue arrive at time 1): the one before will be

$$H(0) = \begin{bmatrix} h_{00}(0) & h_{01}(0) \\ h_{10}(0) & h_{11}(0) \end{bmatrix}$$

and we'd also have

$$H(1) = \begin{bmatrix} h_{00}(1) & 0 \\ 0 & h_{11}(1) \end{bmatrix} :$$



MIMO paths with different delays (blue arrows arrive at time 1)

we have some form of reflection which makes the signal arrive later at the receiver.
Of course if we were to have reflections between 0,1 and 1,0, we'd have

$$H(1) = \begin{bmatrix} h_{00}(1) & h_{01}(1) \\ h_{10}(1) & h_{11}(1) \end{bmatrix}.$$